

Immunodynamic Modeling in Melanoma and Gastric Cancer

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Abstract

In this report, we propose and estimate three immunodynamic ODE models for the effect of upregulation of UCP2 expression on T-cell infiltration in melanoma observed in Cheng et. al 2019 [8]. We then identify immunotherapeutic targets for gastric cancer and do not find a significant differential expression of UCP2, but instead identify a candidate gene, CCR7. We extend the second model to CCR7 and propose it as a novel target for immunotherapy.

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1 Introduction

Cancer is responsible for numerous deaths worldwide, and the latest treatment options for cancer include targeted therapy, immunotherapy, and gene therapy [24]. While there have been significant advancements in clinical trials, issues still need to be addressed to improve cancer therapy. Researchers have been studying the tumour microenvironment (TME) to gain a deeper understanding of cancer, which will help to develop more effective therapeutic designs. Researchers are realizing that cancer actively changes its environment rather than just existing passively inside the body. Re-purposing cellular energy use is one way it accomplishes this, which disrupts the body's defense mechanisms and encourages tumor growth [22]. In this regard, Uncoupling Protein 2 (UCP2), which is present in the inner membrane of mitochondria, has gained notice due to its function in regulating immune and metabolic dynamics [8, 18, 29]. Research indicates that UCP2 may help immune cells' mitochondria function while reducing oxidative stress. The immune system's capacity to identify and eliminate cancerous cells is thought to be enhanced by these alterations, which aids in recruiting T-cells to the TME and maintaining their activity [8].

Mathematical oncology is increasingly being used to study the relationship between immune evasion and tumor metabolism. To measure and forecast T-cell dynamics in solid tumors under various metabolic constraints, a number of biomathematical models have been created [4, 10, 14]. These models demonstrate how T-cell behavior and infiltration are influenced by tumor-induced alterations in energy availability, acidity, and Reactive Oxygen Species (ROS) accumulation. Some researchers use partial differential equations to model the spatial infiltration of immune cells in the tumor mass [10], while Kareva and Berezovskaya’s work [14] describes the reciprocal feedback between tumor metabolism and immune suppression using ODEs. In [17], the authors developed a mathematical framework that depicts the interactions between tumor cells, T cells, and the effects of combination therapy using radiation and anti-PD-L1 immunotherapy, all in a general TME. They also determined crucial parameters that affect tumor control and treatment results through stability and sensitivity analyses. For this study, a modified form of this model will be applied to Melanoma cancer and Gastric cancer.

1.1 Cheng et. al 2019 Summary

Melanoma, a skin cancer originating from the malignancy of melanocytes, has been identified as the fifth most prevalent type of cancer among men and the sixth among women [1]. Melanoma cells possess the ability to modify the self-tolerance pathways of immune checkpoints. They express programmed cell death ligand 1 (PD-L1) and PD-L2, which are ligands for the membrane-bound receptor PD-1. This receptor is found on CD8+ T cells and can be used to prohibit their immune response functions. Consequently, melanoma cells tend to restrict the efficacy of T-cell activation within the tumour. Furthermore, this process also affects B cells and natural killer (NK) cells, as they are alternative PD-1-expressing cells [28]. In Wan-Chen Cheng’s study, the researchers proposed that therapeutic developments targeting UCP2 could be a potential strategy for non-T cell-inflamed tumors in melanoma cancer. This study holds particular relevance in the context of classifying tumors as cold or hot, a framework based on immune cell infiltration and inflammatory activity. Cold tumors, with limited cytotoxic T cell presence and a highly immunosuppressive microenvironment, tend to show poor responses to immunotherapy [5, 7]. In the aforementioned study, they employed a multidimensional scaling process to classify the highly expressed genes associated with defense responses, inflammatory responses, and T cell activation (Figure 1). Based on the expressed genes, UCP2 was identified to be the gene with the highest T-cell activation and correspondingly identified as a potential target for therapeutic intervention [8].

Gastric cancer is the third most common cause of cancer-related deaths [30]. Because Gastric cancer often goes undetected until it is advanced, it continues to be a significant global burden despite advancements in oncology. Its immunosuppressive surrounding cellular environment and treatment resistance exacerbate the problem and make effective intervention challenging [31]. The potential risk factors for gastric cancer include *Helicobacter pylori* infection, dietary habits, genetic predispositions, and age, which are influenced by distinct genetic or environmental factors [20], [12]. Gastric cancers are frequently categorized as cold tumors, which, in contrast to hot tumors, are marked by low antigen presentation, dense stromal barriers, and metabolic conditions that suppress T cell functionality [36]. Meanwhile, the hot tumors contain abundant immune effector cells and are generally more responsive to checkpoint blockade and other immune-targeting strategies. Motivated by these insights, this work aims to identify and characterize an alternative candidate gene that performs comparable metabolic functions to UCP2, but with enhanced potential to decrease tumor burden in gastric cancer (as opposed to melanoma). We aim to investigate how this gene influences tumor size and promotes T-cell recruitment by integrating theoretical modeling with experimental approaches.

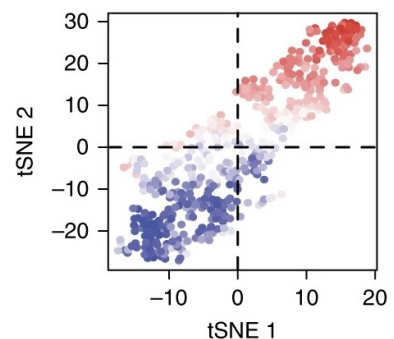


Figure 1: T-distributed stochastic neighborhood embedding (tSNE) plot of melanoma patients according to combined T cell anti-tumor response signature scores.

2 UCP2 Models

The following models are based off of Liao et. al.'s model (2.16) [17] for a general tumor microenvironment. The variables are given below:

$C(t)$: density of tumor cells at time t with unit g/cm^3 ,

$D(t)$: density of DCs at time t with unit g/cm^3 ,

$M_2(t)$: density of M2 macrophages at time t with unit g/cm^3 ,

$T(t)$: density of CD8+ T cells at time t with unit g/cm^3 ,

$T_r(t)$: density of regulatory T cells at time t with unit g/cm^3 ,

$T_\beta(t)$: concentration of TGF- β at time t with unit g/cm^3

$U(t)$: concentration of "effective" UCP2 (UCP2 with doxycycline) at time t with unit g/cm^3

$P(t)$: concentration of PD-1 at time t with unit g/cm^3

$L(t)$: concentration of PD-L1 at time t with unit g/cm^3

$Q(t)$: concentration of the PD-1-PD-L1 complex at time t with unit g^2/cm^6

"Effective" UCP2 refers to UCP2 with doxycycline. This inclusion is motivated by Fig. 3 b) in Cheng et. al. [8] where the UCP2+ tumor only has significantly reduced growth in the presence of doxycycline. The following system network is a modified version of Liao et. al.'s Figure 1 [17] with radiation removed and effective UCP2 added.

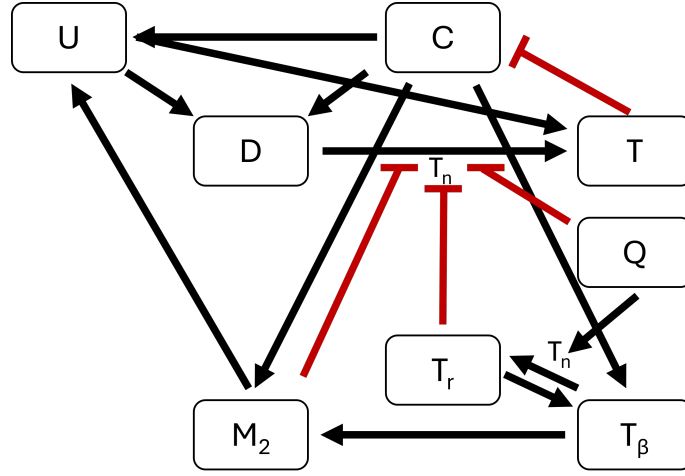


Figure 2: System network for model I. T_n refers to naive T cells.

Figure 3 in Cheng et. al. [8] suggests that the effective UCP2+ TME helps promote the CD8+ T cell concentration. In figure 4 c), the UCP2+ TME sees an increase in dendritic cells. In Table 1 of [16], Li et. al. show that UCP2 expression is higher in melanoma cells than in healthy tissue. The last addition to the network is the production of UCP2 by M_2 which is supported by Fig. 1 g) in Lang et. al. [15].

Model I

$$\left\{ \begin{array}{lcl}
 C' & = & \underbrace{\lambda_C C(1 - \frac{C}{\bar{C}})}_{\text{proliferation}} - \underbrace{\eta_T TC}_{\text{killed by T cell}} - \underbrace{d_C C}_{\text{apoptosis}} \\
 D' & = & \underbrace{\tilde{\lambda}_{DC} \frac{C}{K_C + C}}_{\text{activation by tumor}} + \underbrace{\lambda_{DU} \frac{U}{K_U + U}}_{\text{promotion by UCP2}} - \underbrace{d_D D}_{\text{death}} \\
 M_2' & = & \underbrace{\tilde{\lambda}_{M_2 C} \frac{C}{K_C + C}}_{\text{activation by tumor}} + \underbrace{\lambda_{M_2 T_\beta} M_2 \frac{T_\beta}{K_{T_\beta} + T_\beta}}_{\text{activation by } T_\beta} - \underbrace{d_{M_2} M_2}_{\text{death}} \\
 T' & = & \underbrace{\tilde{\lambda}_T \times \frac{D}{K_D + D}}_{\text{activation by } I_{12}} \times \underbrace{\frac{1}{1 + M_2/\bar{K}_{M_2}}}_{\text{inhibition by } I_{10}} \times \underbrace{\frac{1}{1 + T_r/\bar{K}_{T_r}}}_{\text{inhibition by } T_r} \times \underbrace{\frac{1}{1 + (f_q PL/\bar{K}_Q)}}_{\text{inhibition by Q}} + \underbrace{\lambda_{TU} \frac{U}{K_U + U}}_{\text{recruitment by UCP2}} - \underbrace{d_T T}_{\text{death}} \\
 T_r' & = & \underbrace{\tilde{\lambda}_{T_r T_\beta} \frac{T_\beta}{K_{T_\beta} + T_\beta}}_{\text{activation by } T_\beta} + \underbrace{\tilde{\lambda}_{T_r Q} \frac{f_q PL}{K_Q + f_q PL}}_{\text{activation by Q}} - \underbrace{d_{T_r} T_r}_{\text{death}} \\
 T_\beta' & = & \underbrace{\lambda_{T_\beta C} C}_{\text{production by } C} + \underbrace{\lambda_{T_\beta T_r} T_r}_{\text{production by } T_r} - \underbrace{d_{T_\beta} T_\beta}_{\text{degradation}} \\
 U' & = & \underbrace{f_{dox}(t)}_{\text{dox injection}} \left(\underbrace{\lambda_{UC} \frac{C}{K_C + C}}_{\text{Production by tumor}} + \underbrace{\lambda_{UM_2} \frac{M_2}{K_{M_2} + M_2}}_{\text{Production by M2 macrophages}} \right) - \underbrace{d_U U}_{\text{death}}
 \end{array} \right. \quad (1)$$

with

$$P = \rho_P T, \quad L = \rho_L (T + \epsilon M_2 + \epsilon C)(1 + \lambda_{LU} U), \quad Q = P \cdot L \quad (2)$$

and $f_q \in [0, 1]$ indicates the anti-PD-L1/anti-PD-1 efficacy with 1 being no treatment and 0 being maximal treatment (complete removal of the Q complex). $f_{dox}(t)$ is the doxycycline injection. This multiplies the production terms, so the efficacy of the UCP2 will be based on the presence of doxycycline. A point of further investigation is to ascertain to what extent the concentration of doxycycline makes the UCP2 “effective” in the system as the mechanism may differ from here.

Model II is a reduction of model I in which the T_r and T_β variables and corresponding equations are removed.

Model II

$$\left\{ \begin{array}{lcl}
 C' & = & \underbrace{\lambda_C C(1 - \frac{C}{\bar{C}})}_{\text{proliferation}} - \underbrace{\eta_T TC}_{\text{killed by T cell}} - \underbrace{d_C C}_{\text{apoptosis}} \\
 D' & = & \underbrace{\tilde{\lambda}_{DC} \frac{C}{K_C + C}}_{\text{activation by tumor}} + \underbrace{\lambda_{DU} \frac{U}{K_U + U}}_{\text{promotion by UCP2}} - \underbrace{d_D D}_{\text{death}} \\
 M_2' & = & \underbrace{\tilde{\lambda}_{M_2 C} \frac{C}{K_C + C}}_{\text{activation by tumor}} + \underbrace{\tilde{\lambda}_{M_2 Q} M_2 \times (1 + \frac{f_q PL}{K_Q + f_q PL})}_{\text{activation by Q}} - \underbrace{d_{M_2} M_2}_{\text{death}} \\
 T' & = & \underbrace{\tilde{\lambda}_T \times \frac{D}{K_D + D}}_{\text{activation by } I_{12}} \times \underbrace{\frac{1}{1 + M_2/\bar{K}_{M_2}}}_{\text{inhibition by } I_{10}} \times \underbrace{\frac{1}{1 + (f_q PL/\bar{K}_Q)}}_{\text{inhibition by Q}} + \underbrace{\lambda_{TU} \frac{U}{K_U + U}}_{\text{recruitment by UCP2}} - \underbrace{d_T T}_{\text{death}} \\
 U' & = & \underbrace{f_{dox}(t)}_{\text{dox injection}} \left(\underbrace{\lambda_{UC} \frac{C}{K_C + C}}_{\text{Production by tumor}} + \underbrace{\lambda_{UM_2} \frac{M_2}{K_{M_2} + M_2}}_{\text{Production by M2 macrophages}} \right) - \underbrace{d_U U}_{\text{death}}
 \end{array} \right. \quad (3)$$

with

$$P = \rho_P T, \quad L = \rho_L (T + \epsilon M_2 + \epsilon C)(1 + \lambda_{LU} U), \quad Q = P \cdot L. \quad (4)$$

We reduce the model further into model III by removing D and M_2 from the model.

Model III

$$\left\{ \begin{array}{l} C' = \underbrace{\lambda_C C (1 - \frac{C}{\bar{C}})}_{\text{proliferation}} - \underbrace{\eta_T T C}_{\text{killed by T cell}} - \underbrace{d_C C}_{\text{apoptosis}} \\ T' = \underbrace{\tilde{\lambda}_T \times \frac{C}{K_C + C}}_{\text{activation by } I_{12}} \times \underbrace{\frac{1}{1 + (f_q P L / \bar{K}_Q)}}_{\text{inhibition by Q}} + \underbrace{\lambda_{TU} \frac{U}{K_U + U}}_{\text{recruitment by UCP2}} - \underbrace{d_T T}_{\text{death}} \\ U' = \underbrace{f_{dox}(t)}_{\text{dox injection}} \left(\underbrace{\lambda_{UC} \frac{C}{K_C + C}}_{\text{Production by tumor}} + \underbrace{\lambda_{UQ} \frac{f_q P L}{K_Q + f_q P L}}_{\text{Production by M2 macrophages}} \right) - \underbrace{d_U U}_{\text{death}} \end{array} \right. \quad (5)$$

with

$$P = \rho_P T, \quad L = \rho_L (T + \epsilon C) (1 + \lambda_{LU} U), \quad Q = P \cdot L. \quad (6)$$

3 Parameter Information

3.1 Parameter Definitions

Table 1: UCP2 models parameter definitions.

Parameter	Definition	Model(s) Used In
λ_C	Growth rate of C	I-III
$\tilde{C}$	Tumor carrying capacity	I-III
η_T	Killing rate of C by T	I-III
d_C	Death rate by C	I-III
λ_{DC}	Activation rate of D by C	I, II
D_0	Average density of immature DCs	I, II
K_C	Half-saturation of tumor	I-III
λ_{DU}	Promotion rate of D by U	I, II
K_U	Half-saturation of UCP2	I-III
d_D	Death rate of D	I, II
λ_{M_2C}	Promotion rate of M_2 by C	I, II
M_0	Average concentration of monocytes	I, II
$\lambda_{M_2T_\beta}$	Promotion rate of M_2 by T_β	I
λ_{M_2Q}	Promotion rate of M_2 by Q	II
K_{T_β}	Half saturation of TGF- β	I
d_{M_2}	Death rate of M_2	I, II
λ_T	Activation rate of T by IL-12	I-III
T_n	Average concentration of naive T cells	I-III
K_D	Half saturation of DCs	I, II
$\bar{K}_{M_2}$	Inhibition on T by IL-10	I, II
$\bar{K}_{T_r}$	Inhibition on T by T_r	I
$\bar{K}_Q$	Inhibition on T by Q	I-III
λ_{TU}	Rate of recruitment of T by UCP2	I-III
d_T	Death rate of T	I-III
$\lambda_{T_rT_\beta}$	Promotion rate of T_r by T_β	I
λ_{T_rQ}	Promotion rate of T_r by Q	I
K_Q	Half-saturation of Q	I-III
d_{T_r}	Death rate of T_r	I
$\lambda_{T_\beta C}$	Promotion rate of T_β by C	I
$\lambda_{T_\beta T_r}$	Promotion rate of T_β by T_r	I
d_{T_β}	Death rate of T_β	I
λ_{UC}	Production rate of U by C	I-III
λ_{UM_2}	Production rate of U by M_2	I, II
λ_{UQ}	Production rate of U by M_2 through Q	III
K_{M_2}	Half saturation of M_2	I-II
d_U	Degradation rate of U	I-III
ρ_P	Mass ratio of P to T	I-III
ρ_L	Mass ratio of L to T	I-III
ϵ	Expression rate of L on C	I-III
λ_{LU}	Promotion rate of L by U	I-III

Table 2: UCP2 models combination parameters.

Parameter	Formula
$\tilde{\lambda}_{DC}$	$\lambda_{DC} \cdot D_0$
$\tilde{\lambda}_{M_2C}$	$\lambda_{M_2C} \cdot M_0$
$\tilde{\lambda}_{M_2Q}$	$\lambda_{M_2Q} \cdot M_0$
$\tilde{\lambda}_T$	$\lambda_T \cdot T_n$
$\tilde{\lambda}_{T_rT_\beta}$	$\lambda_{T_rT_\beta} \cdot T_n$
$\tilde{\lambda}_{T_rQ}$	$\lambda_{T_rQ} \cdot T_n$

3.2 Parameter Estimation

For the parameter estimations in table 3, all values not calculated in this section are taken from Table 1 in Liao et. al. [17]. Note that f_q is taken to be 1 as the estimations do not include any anti-PD-1/PD-L1 treatments. Also, $f_{dox}(t)$ will be set to the treatment setting of 1.

Estimation of the steady state for C

In Liao et. al.'s paper, the steady state of C (C^*) is estimated to be 0.4 g/cm^3 . To estimate the new steady state we consider Fig. 3 b) in Cheng et. al. [8] and take the ratio of the 3F-UCP2+Dox tumor to the 3F (UCP2-) tumor, which is 0.1728, and apply it to the previous steady state yielding $C^* = 0.1728 \cdot 0.4 \text{ g/cm}^3 = 0.0691 \text{ g/cm}^3$.

Estimation of the steady state for T

We use Fig. 3 e) in Cheng et. al. to find a ratio of T cells in the UCP+ vs UCP- environment and obtain a value of 3.857. We apply this ratio to the previous steady state in Liao. et. al. of $5 \times 10^{-4} \text{ g/cm}^3$ to obtain a new steady state $T^* = 3.857 \cdot 5 \times 10^{-4} \text{ g/cm}^3 = 0.001929 \text{ g/cm}^3$.

Estimation of the steady state for D

We use Fig. 4 c) in Cheng et. al. to find a ratio of T cells in the UCP+ vs UCP- environment and obtain a value of 4.041. We apply this ratio to the previous steady state in Liao et. al. of 4×10^{-5} to obtain a new steady state $D^* = 1.6164 \times 10^{-4} \text{ g/cm}^3$.

Estimation of the steady state for U

Noting that UCP2 is a mitochondrial membrane, we make the following assumptions to obtain an estimation for the steady state of U . First, we assume that, since mitochondria are highly folded, that the membrane makes up the entire volume of the mitochondria. Next we note that around 75% of the mitochondria's membrane's mass is protein [2]. In order to use the claim that 0.01% to 0.1% of a cell's membrane proteins are UCP2 [6], we assume that the membrane proteins are uniformly distributed across all membranes in the cell and use a median value of 0.05%. Applying this to the average mitochondria concentration in a cell of 1.18 g/cm^3 [23], yields the following computation

$$U^* = 0.75 \cdot 0.0005 \cdot 1.18 \text{ g/cm}^3 = 4.425 \times 10^{-4} \text{ g/cm}^3.$$

Estimation of λ_{LU} and L^*

Using Figure 1 g) in Cheng et. al. we calculate the ratio between the UCP2+ PD-L1 expression and UCP2- PD-L1 expression to be 0.2333. We then assume that this represents the ratio between the wild-type (using the value from Liao et. al.'s paper of $4.6983 \times 10^{-10} \text{ g/cm}^3$) and the UCP2+ case (L^*) which yields

$$\frac{4.6983 \times 10^{-10} \text{ g/cm}^3}{L^*} = 0.2333.$$

Solving for L^* yields $2.014 \times 10^{-9} \text{ g/cm}^3$. Explicitly, for model 2, writing the expressions for the wild-type L and the UCP2+ L and equating to the same ratio allows us to solve for λ_{LU} :

$$\frac{\rho_L(T_{WT}^* + \epsilon M_{2WT}^* + \epsilon C_{WT}^*)}{\rho_L(T^* + \epsilon M_2^* + \epsilon C^*)(1 + \lambda_{LU}U^*)} = \frac{5 \times 10^{-4} + 10^{-3} \cdot 5.73 \times 10^{-5} + 10^{-3} \cdot 0.4}{(0.001929 + 10^{-3} \cdot 5.73 \times 10^{-5} + 10^{-3} \cdot 0.1031)(1 + \lambda_{LU} \cdot 4.425 \times 10^{-4})} = 0.2333$$

and thus $\lambda_{LU} = 2030.385 \text{ cm}^3/\text{g}$. For model 3, we calculate

$$\frac{\rho_L(T_{WT}^* + \epsilon C_{WT}^*)}{\rho_L(T^* + \epsilon C^*)(1 + \lambda_{LU}U^*)} = \frac{5 \times 10^{-4} + 10^{-3} \cdot 0.4}{(0.001929 + 10^{-3} \cdot 0.1031)(1 + \lambda_{LU} \cdot 4.425 \times 10^{-4})} = 0.2333.$$

This yields $\lambda_{LU} = 2030.233 \text{ g/cm}^3$ for model 3.

Estimation of K_U

At equilibrium, the half saturation value of K_U can be approximated as its steady state leading to $K_U = 4.425 \times 10^{-4} \text{ g/cm}^3$.

Estimation of λ_{TU}

Since the T cell expression increases nearly 4 fold in the UCP2+ case as compared to the wild type (see the ratio calculated in the estimation of the steady state for T section) we assume that $\lambda_{TU} = 2\lambda_T = 2 \cdot 48.6 \times 10^{-4} = 97.2 \text{ g/(cm}^3 \cdot \text{day)}$

Estimation of d_U

In Rousett et. al. [25], the half life of UCP2 is given as 30 minutes (1/48 of a day). Solving the half life equation

$$\frac{1}{2} = e^{-d_U \frac{1}{48}}$$

yields $d_U = 33.271 \text{ day}^{-1}$.

Estimation of $\tilde{\lambda}_{M_2Q}$

In model II, we estimate $\tilde{\lambda}_{M_2Q}$ by substituting the other known values into the M_2 equation and isolating for $\tilde{\lambda}_{M_2Q}$ which yields 0.0069 day^{-1} .

Estimation of λ_{UC} , λ_{UM_2} , λ_{UQ}

For model II, we assume that UCP2 is produced more by tumor cells than by macrophages and take $\lambda_{UC} = 10\lambda_{UM_2}$. Substituting into the U equation:

$$0 = \frac{0.0691}{0.4 + 0.0691} 10\lambda_{UM_2} + \frac{1}{2}\lambda_{UM_2} - 33.271 \cdot 4.425 \times 10^{-4}$$

which yields $\lambda_{UM_2} = 0.0075 \text{ day}^{-1}$ and $\lambda_{UC} = 0.075 \text{ day}^{-1}$. For model 3, since the λ_{UQ} term is the indirect production of U through Q promoting M_2 , we assume that $\lambda_{UC} = 100\lambda_{UQ}$. Substituting into the U equation yields

$$0 = \frac{0.0691}{0.4 + 0.0691} 100\lambda_{UQ} + \frac{1}{2}\lambda_{UQ} - 33.271 \cdot 4.425 \times 10^{-4}$$

which yields $\lambda_{UQ} = 9.667 \times 10^{-4} \text{ day}^{-1}$ and $\lambda_{UC} = 0.0967 \text{ day}^{-1}$.

Estimation of λ_{DU}

The base value for this was adjusted manually in order to capture the behaviour of the dendritic cells in the data, increasing under UCP2 + DOX and we found $5 \times 10^{-4} \text{ g}/(\text{cm}^3 \cdot \text{day})$ to be an effective estimation.

Table 3: **Parameter estimation for UCP2 model II.**

Parameter	Estimated Value	Parameter	Estimated Value
λ_C	0.5436 day^{-1}	K_D	$4 \times 10^{-5} \text{ g}/\text{cm}^3$
$\tilde{C}$	$0.9 \text{ g}/\text{cm}^3$	$\bar{K}_{M_2}$	$2.87 \times 10^{-5} \text{ g}/\text{cm}^3$
η_T	$24 \text{ cm}^3/(\text{g} \cdot \text{day})$	$\bar{K}_Q$	$2.9247 \times 10^{-20} \text{ g}^2/\text{cm}^6$
d_C	0.17 day^{-1}	λ_{TU}	$97.2 \text{ g}/(\text{cm}^3 \cdot \text{day})$
$\tilde{\lambda}_{DC}$	$8 \times 10^{-6} \text{ g}/(\text{cm}^3 \cdot \text{day})$	d_T	0.18 day^{-1}
K_C	$0.4 \text{ g}/\text{cm}^3$	λ_{UC}	0.075 day^{-1}
λ_{DU}	$5 \times 10^{-4} \text{ g}/(\text{cm}^3 \cdot \text{day})$	λ_{UM_2}	0.0075 day^{-1}
K_U	$4.425 \times 10^{-4} \text{ g}/\text{cm}^3$	K_{M_2}	$5.73 \times 10^{-5} \text{ g}/\text{cm}^3$
d_D	0.1 day^{-1}	d_U	33.271 day^{-1}
$\tilde{\lambda}_{M_2C}$	$1.0016 \times 10^{-6} \text{ g}/\text{cm}^3$	ρ_P	2.49×10^{-7}
$\tilde{\lambda}_{M_2Q}$	0.0069 day^{-1}	ρ_L	5.22×10^{-7}
K_Q	$5.8494 \times 10^{-20} \text{ g}^2/\text{cm}^6$	ϵ	10^{-3}
d_{M_2}	0.015 day^{-1}	λ_{LU}	$2030.385 \text{ g}/\text{cm}^3$
$\tilde{\lambda}_T$	$48.6 \times 10^{-4} \text{ g}/(\text{cm}^3 \cdot \text{day})$		

Table 4: **Parameter estimation for UCP2 model III .**

Parameter	Estimated Value	Parameter	Estimated Value
λ_C	0.5436 day^{-1}	d_T	0.18 day^{-1}
$\tilde{C}$	$0.9 \text{ g}/\text{cm}^3$	λ_{UC}	0.0967 day^{-1}
η_T	$24 \text{ cm}^3/(\text{g} \cdot \text{day})$	λ_{UQ}	$9.667 \times 10^{-4} \text{ day}^{-1}$
d_C	0.17 day^{-1}	K_Q	$5.8494 \times 10^{-20} \text{ g}^2/\text{cm}^6$
$\tilde{\lambda}_T$	$48.6 \times 10^{-4} \text{ g}/(\text{cm}^3 \cdot \text{day})$	d_U	33.271 day^{-1}
K_C	$0.4 \text{ g}/\text{cm}^3$	ρ_P	2.49×10^{-7}
$\bar{K}_Q$	$2.9247 \times 10^{-20} \text{ g}^2/\text{cm}^6$	ρ_L	5.22×10^{-7}
λ_{TU}	$97.2 \text{ g}/(\text{cm}^3 \cdot \text{day})$	ϵ	10^{-3}
K_U	$4.425 \times 10^{-4} \text{ g}/\text{cm}^3$	λ_{LU}	$2030.233 \text{ g}/\text{cm}^3$

3.3 Mathematical analysis of models II and III

Remark 1. By substituting the parameter values from Table 3 into model II and solving the system at equilibrium, we find;

1. *In the wild-type case (i.e. when UCP2 is not considered), model II without U admits a total of 10 fixed points where only two of them are relevant. The first fixed point,*

$$E^* = \{C^* = 0.5955, D^* = 0.0000478554, M_2^* = 0.000151925, T^* = 0.00057991\}$$

is stable with eigenvalues of the Jacobian matrix given by:

$$\{-33.2718, -0.088695 + 0.139148 \text{ Im}, -0.088695 - 0.139148 \text{ Im}, -0.1, -0.00299452\}.$$

The remaining fixed point is a tumor free equilibrium, $E^0 = \{C = M_2 = T = 0\}$, and it is unstable with eigenvalues

$$\{0.3736, -0.18, -0.1, -0.0081\}.$$

2. *In the UCP2+ case (i.e. when UCP2 is considered), model II admits a total of 34 fixed points. Among these, there exists one boundary equilibrium given by $E^0 = \{C = D = M_2 = T = U = 0\}$ which is unstable since the eigenvalues of the Jacobian at this point are given by $\{-33.271, 0.3736, -0.18, -0.1, -0.0081\}$. The other relevant equilibrium point is given by*

$$E^* = \{C^* = 2.59781 \times 10^{-6}, D^* = 1.44654 \times 10^{-7}, M_2^* = 5.36355 \times 10^{-9}, T^* = 0.0155666, U^* = 1.27563 \times 10^{-8}\}. \quad (7)$$

At this point, the Jacobian matrix has a characteristic polynomial with factors:

$$(\phi + 0.00299503)(\phi + 0.100001)(\phi + 33.2718)(\phi^2 + 0.177388\phi + 0.0272277),$$

yielding eigenvalues:

$$\{-33.2718, -0.0886942 + 0.139144 \text{ Im}, -0.0886942 - 0.139144 \text{ Im}, -0.100001, -0.00299503\}.$$

Since all the eigenvalues have negative real parts, this fixed point is locally asymptotically stable.

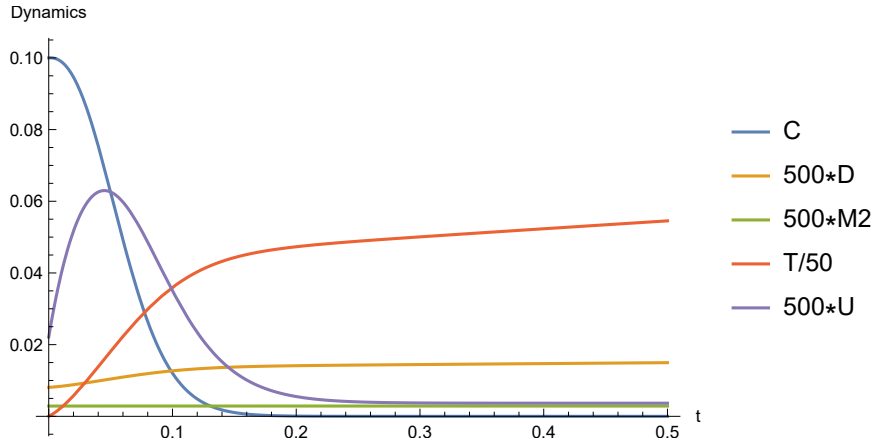


Figure 3: Time plot of the dynamics corresponding to Model II in the presence of UCP2.

Remark 2. *Note that the inclusion of the UCP2 into our model II has a crucial impact on tumor dynamics. Compared to the wild-type case, when this gene is not considered, the value of the cancer tumor at equilibrium decreased significantly when UCP2 is active. Biologically, this aligns with recent findings suggesting that UCP2 plays a regulatory role in immune response, cellular metabolism, and oxidative stress.*

From a modeling perspective, the observed decrease in tumor cell counts upon UCP2 inclusion suggests that UCP2 acts as a tumor-suppressive factor within the modeled system.

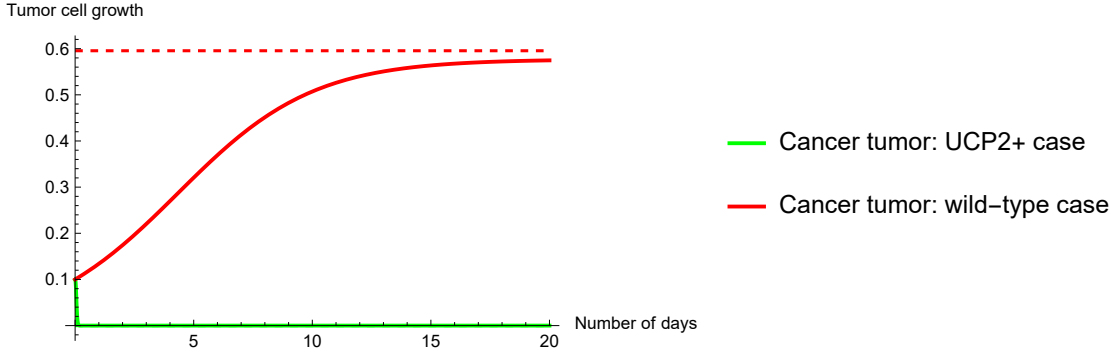


Figure 4: The plot of tumor cell growth in the wild-type and UCP2+ cases with time.

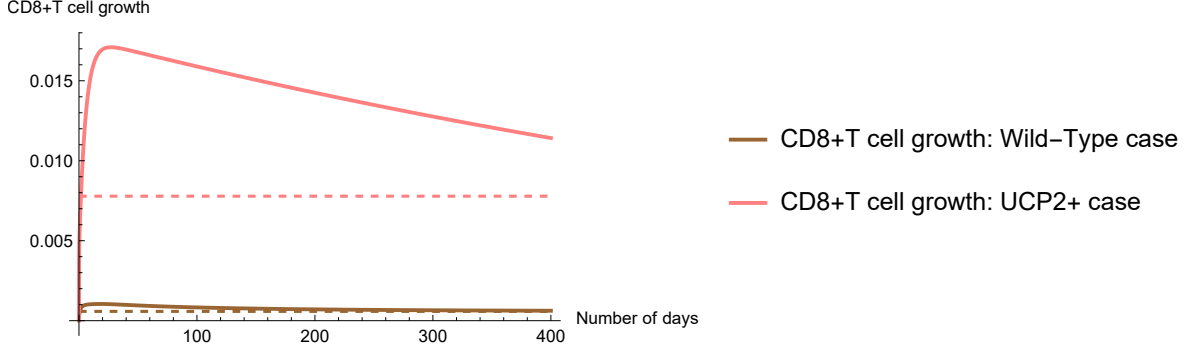


Figure 5: The plot of T-cells growth in the wild-type and UCP2+ cases with time.

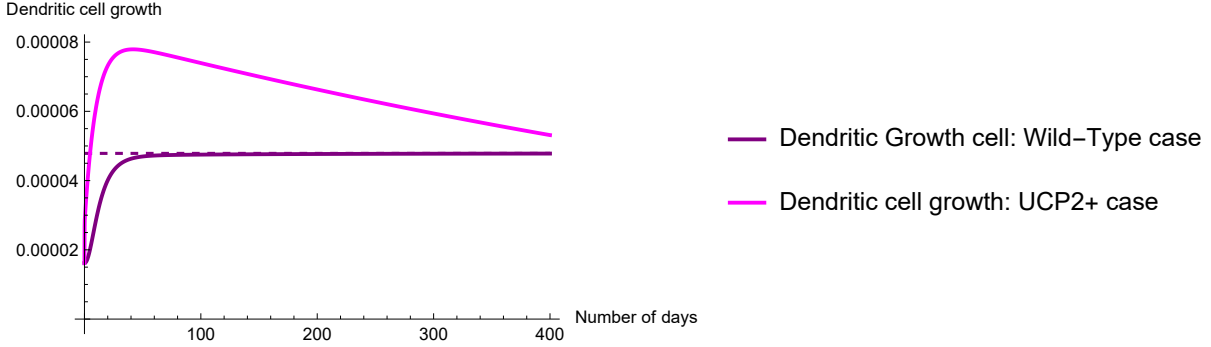


Figure 6: The plot of dendritic cell growth in the wild-type and UCP2+ cases with time.

Biological Interpretation.

The equilibrium point (7) corresponds to a biologically feasible steady state of the tumor-immune interaction model II. Indeed, at this point, the value of the tumor C^* is extremely low, suggesting effective control or near elimination of the cancer cells. The CD8+T cells (with equilibrium T^*) remain relatively high, indicating a strong and sustained cytotoxic immune response. This implies that the immune system, in particular through cytotoxic T lymphocytes, is able to keep the tumour under control. The level of dendritic cells D^* and macrophages M_2^* are both very low, which may reflect a limited role in their maintenance of the TME compared with the CD8+ T cells. The UCP2 gene value U^* is also low. Considering figure 3 in tandem with this, this seems to suggest that once T cells have sufficiently infiltrated the microenvironment, the levels of UCP2 decrease accordingly.

Altogether, E^* indicates a tumor controlled, predominantly immune state (or almost tumor free equilibrium); the tumor has not been completely eradicated, but its growth is effectively suppressed by a robust CD8+T cell presence. The low level of M_2 typically associated with immuno elimination and tumor progression enhances the idea that the TME is not favoring immune escape mechanisms. Moreover, the local stability of this equilibrium implies that under the current parameter values, small perturbations in the system (e.g. fluctuations in cell populations or gene expression) will not destabilize this favorable state.

Remark 3. Alternatively, in the case of model III, we substitute the parameter values from Table 4 into model III and by solving the system at the equilibrium, we find;

1. **In the wild-type case**, the model (5) without U admits five fixed points; a boundary one ($C = T = 0$) with corresponding eigenvalues of the Jacobian $(0.3736, -0.18)$ yielding to its unstability and a positive equilibrium point ($C^* = 0.536441, T^* = 0.00206625$), which is locally stable with eigenvalues of the Jacobian at this point $(-0.487839, -0.314137)$.
2. **In the UCP2+ case**, the model (5) admits a total of 13 fixed points where only three of them are relevant:

$$\begin{cases} E_0 = (C = T = U = 0), \\ E_1 = (c = 0, T = 4.20909 * 10^{-8}, U = 3.44912 * 10^{-14}), \\ E_2 = (c = 0, T = 10.4843, U = 8.76138 * 10^{-6}), \end{cases}$$

where the third point is the unique stable point since the corresponding eigenvalues of the Jacobian are $(-251.249, -33.271, -0.18)$.

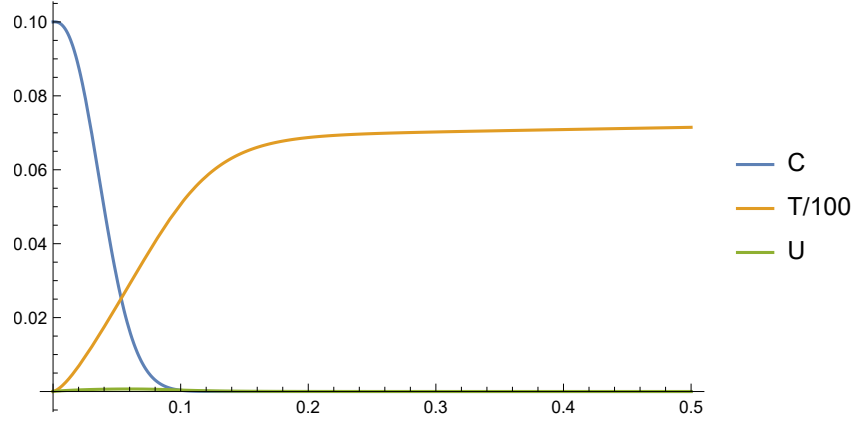


Figure 7: Time plot of the dynamics of the model (5) when UCP2 is included.

4 Gastric Cancer Analysis

In this section, we report on our findings from Differentially Expressed Gene (DEG) analysis on the TCGA Stomach Adrenocarcinoma (STAD) mRNA expression dataset, following Cheng et al.'s methodology [8], and justify our choice of candidate gene with biological and statistical evidence.

4.1 Hot-Cold Tumor Classification

The hot and cold tumor phenotype is defined in the literature heuristically i.e, there is no canonical gene set; in this paper, we use the seven immune response gene sets defined in Cheng et. al 2019 [8] to characterize the immune response. To compute the enrichment scores of these gene sets, we use the Single-sample Gene Set Enrichment Analysis (ssGSEA) projection methodology from Barbie et. al 2009 [3]. Hot-cold tumor immune response enrichment score clustering is visualized with tSNE in Figure 8a along with differential gene expression in 8b. We see two distinct clusters emerge between hot and cold tumors in tSNE indicating good representation performance of the selected genesets. Moreover, in Figure 8b, there appears to be a symmetric distribution of differential expression suggesting that transcriptional differences in hot versus cold tumors is gene-specific.

4.2 Candidate Gene Selection

In selecting immunotherapy targets for gastric cancer, we consider a few criteria to identify candidate genes that work well with our existing models. The candidate gene should (1) be significantly differentially expressed between hot and cold tumors, (2) be expressed by the tumor and M_2 macrophages, and (3) induce T cell and dendritic cell activity through its expression. We find around 2,000 genes that were significantly differentially expressed after false discovery rate correction. Notably, UCP2 was not among the significantly differentially expressed genes (DEGs). We note the most significantly DEGs in Table 5; these genes were either incident to the immune response rather than modulatory and/or did not fit our modelling criteria. We further investigate CCR7 as it best fits our modelling criteria.

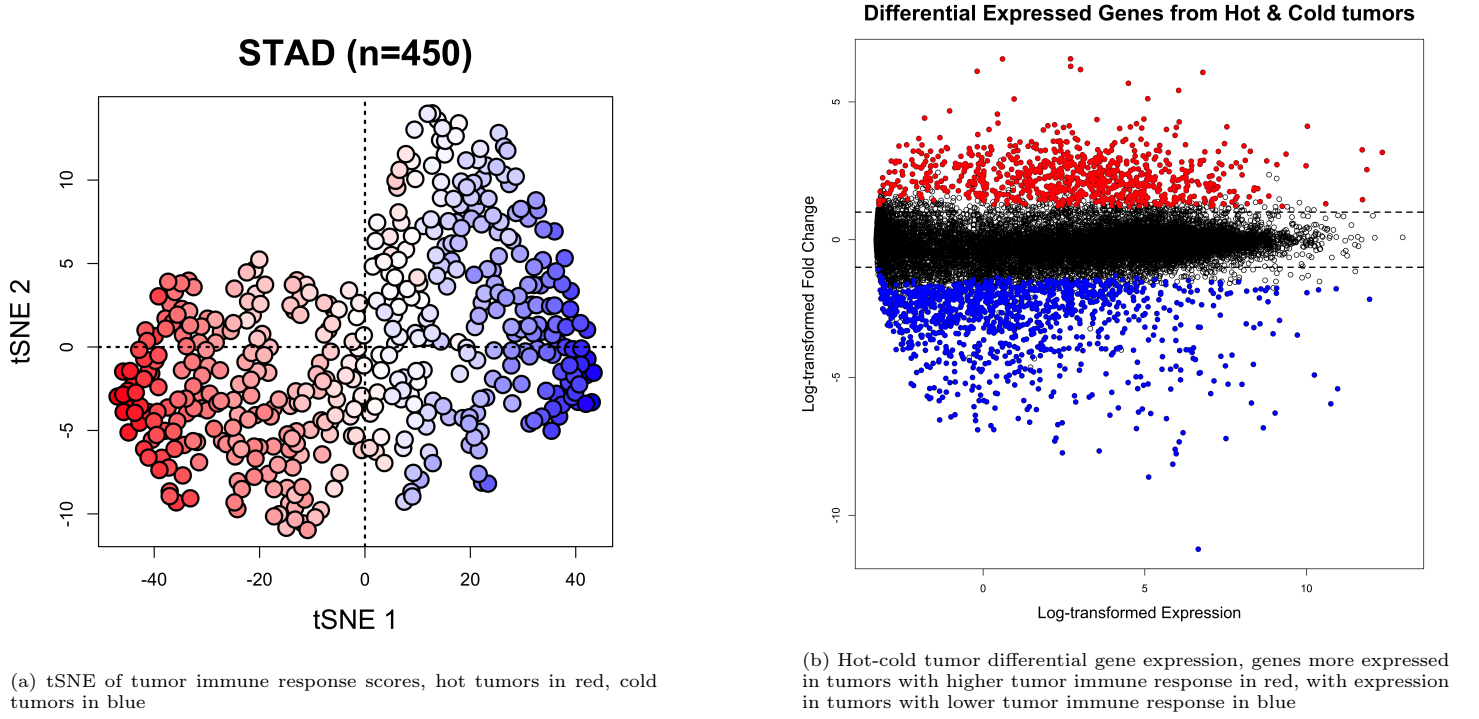


Figure 8: Hot-cold tumor classification

Table 5: Top DEGs by FDR-corrected pValue

Rank	Gene	logFC	FDR-corrected pValue
1	IDO1	6.07158265	1.82842866352211e-79
2	CCL5	3.44753797	4.70780699079878e-73
3	NKG7	3.56753372	1.32594350580427e-72
4	GBP5	4.07761233	2.60797483990634e-67
5	GBP4	3.48675737	4.85687299380203e-61
6	CD48	3.22600646	1.15221780554412e-60
7	CXCR2P1	4.13497154	1.3593733424391e-58
8	CD8A	3.42750636	4.30535963076434e-58
9	CCR5	2.95129634	5.15843173839676e-58
10	NCF1	3.16798647	4.74249244551847e-57
...	...	...	...
248	CCR7	3.07829456	1.73656602177637e-22

4.3 Biological Basis for CCR7

According to [8], UCP2 was identified as a potential targeted gene for therapeutic advancements in melanoma cancer. Furthermore, this study found that UCP2 increases CD8⁺ T cell and dendritic cell infiltration into the TME. By providing doxycycline to UCP2 expressing melanoma cell lines, it was demonstrated that UCP2 over-expression in these tumour cells enhances the immune environment. UCP2 facilitated the recruitment of CD8⁺ T cells by releasing chemokines, combined with a decrease in the expression of pro-tumorigenic factors.

CCR7 is a chemokine receptor produced by the CCR7 gene. It is renowned for its capacity to stimulate cell migration. However, its original classification was attributed to its ability to direct the infiltration of inflammatory cells [13,19]. Exploring the migration response of CCR7 protein-coding T cells within specific chemokine environments will facilitate a comprehensive understanding of the T-cell migration process [33]. Evidence suggests that the expression of CCR7 in gastric cancer tumour tissue is 52.9%. Furthermore, an increase in tumor severity correlates with an increase in the expression of CCR7 [32]. We use this as evidence for the criteria that CCR7 be produced by the tumor, just as UCP2 is. The increased CCR7 expression also enables dendritic cells to migrate towards the lymph nodes [27], which enhances the promotion of dendritic cells. CCR7 expression regulation can activate and migrate CD8⁺ T cells into the tumour cells [9], resembling functions similar to UCP2. In [37], they investigated the role of CCR7 in promoting macrophage recruitment in oral squamous cell carcinoma and confirmed a correlation between CCR7 expression and increased infiltration of M2 macrophages. However, there was no

exploration towards establishing a link between CCR7 and the promotion of M2 macrophages in gastric cancer. However, based on the current values of the existing model, the difference in the UCP2 production with or without the macrophages is minimal (Figure 9). Thus, with the correct parameters, the model can capture the case where the M2 macrophages make a minimal difference on the value of U in melanoma as well as in gastric cancer.

Nevertheless, the gene CCR7 presents itself as a potential therapeutic target for cancer in two distinct ways. Firstly, its functional capability to facilitate the trafficking of cells involved in the production of an immune response is noteworthy. Secondly, it also controls the migration of tumour cells towards the lymph nodes. Given the extensive literature available on CCR7, it is evident that CCR7 shares comparable functionalities with UCP2 with the potential exception of the M2 macrophages.

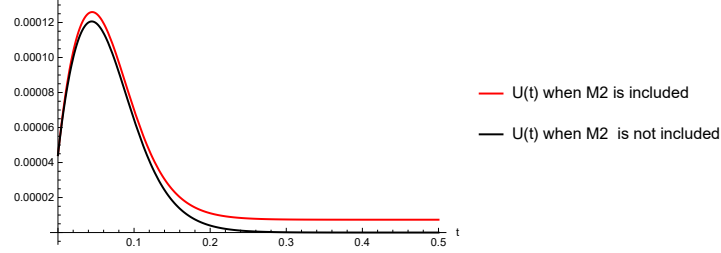


Figure 9: Time plot of UCP2 dynamics in the cases when M_2 macrophages are included and excluded in U'

4.4 Immunogenic Characterization of CCR7

Previous studies [32, 35] have used immunohistological and experimental methods to investigate CCR7's role in cancer; here we leverage high-throughput mRNA sequence and masked somatic mutation data derived from simple nucleotide variation (SNV) to determine immunogenic correlates of CCR7 expression. CCR7 appears to be highly expressed in the hot tumor phenotype (Figure 10a). In Figure 10b, we see that CCR7 is consistently co-expressed with individual genes related to T cell infiltration and anti-tumor immunity. Whether this is an artifact of inflammation is unclear, but the strong correlations observed in Figure 10b suggest a key immunogenic role for CCR7.

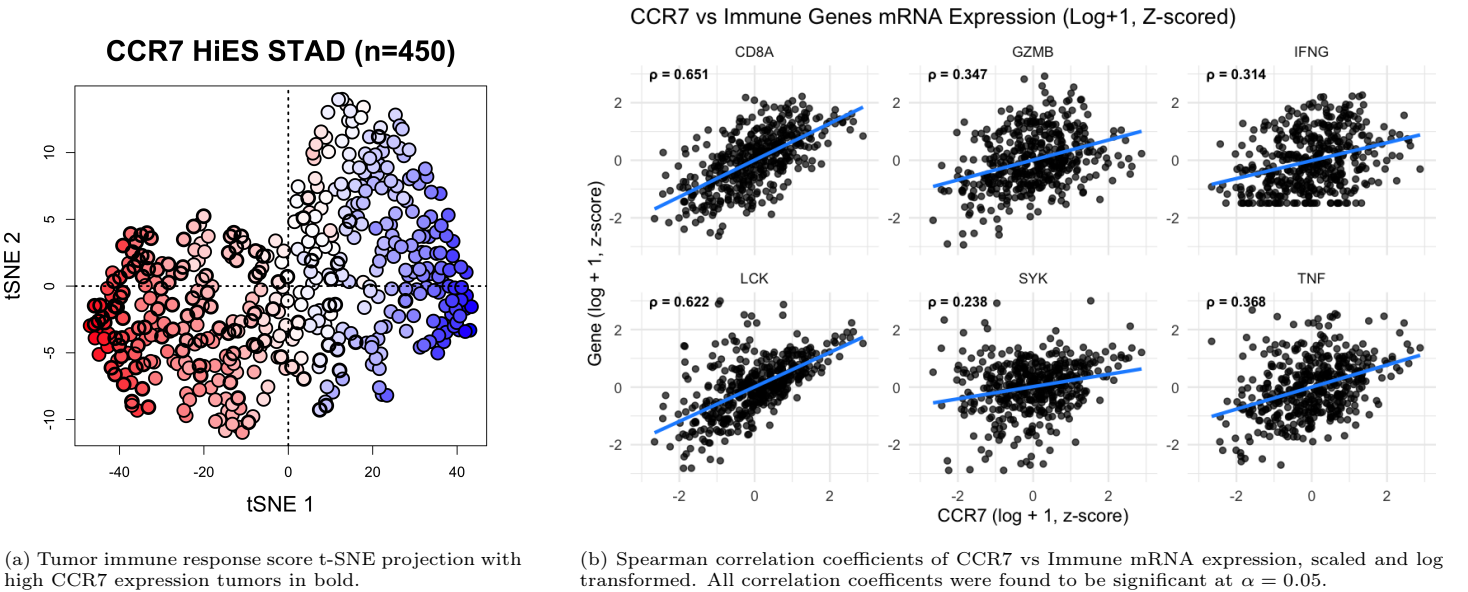


Figure 10: Correlates of CCR7 and Immunogenic Activity

Epigenetic regulator genes are "meta-genes" that control chromatin organization and gene regulation processes. Mutations in these genes can systemically alter gene expression patterns. Non-silent mutations in epigenetic regulator genes have been associated with epigenetic instability, and several particularly aggressive cancers exhibit high rates of these mutations [26]. Tumors with a high expression of CCR7 (HiES) had significantly more epigenetic gene mutations compared to tumors with a low expression of CCR7 (LoES) suggesting a potential relationship between CCR7 and tumor epigenetic instability and the development of aggressive, undifferentiated tumors (Figure 11a). In particular, there is a significant difference between HiES and LoES tumors in KMT2D, an epigenetic regulator gene, mutations (Figure 11b) which have been implicated in better

response to immune checkpoint inhibitors (ICI) therapy and increased immunogenicity [34]. KMT2D mutations seem to be primarily observed in hot tumors (Figure 11c) in gastric cancer, a result consistent with Wang et. al 2024. This suggests a potentially useful mechanism of action for CCR7-based immunotherapy.

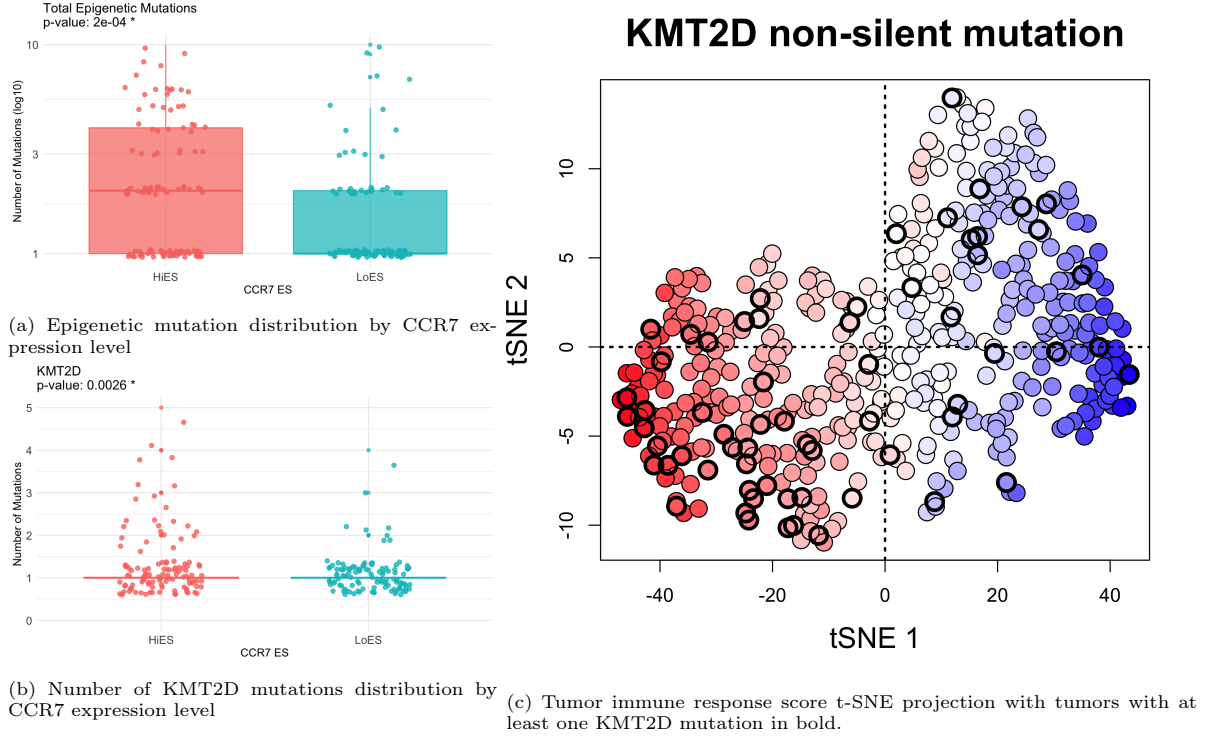


Figure 11: Non-silent epigenetic gene mutation incidence and transcriptomic clustering in tumors with high and low expression of CCR7 defined as 75th and 25th percentile of CCR7 mRNA expression levels. pValues computed with a two-sample t-test with $*$ indicating significance at $\alpha = 0.05$.

4.5 Re-interpretation of the UCP2 Melanoma Model to the CCR7 STAD Setting

Since we have established that CCR7 behaves comparably to UCP2, we propose applying model 2 and 3 from the melanoma setting to the gastric cancer setting. The models' non-UCP2 interactions are based off of Liao et. al.'s general TME model [17] and thus are reasonable to include into the gastric cancer TME. We reinterpret the U variable as concentration of CCR7 at time t with unit g/cm^3 and the parameter definitions change accordingly:

Table 6: Modified parameter definitions for CCR7 Model adaptations

Parameter	Definition
λ_{DU}	Promotion rate of D by CCR7
K_U	Half-saturation of CCR7
λ_{TU}	Rate of recruitment of T by CCR7
λ_{UC}	Production rate of CCR7 by C
λ_{UM_2}	Production rate of CCR7 by M_2
λ_{UQ}	Production rate of CCR7 by M_2 through Q
d_U	Degradation rate of CCR7
λ_{LU}	Promotion rate of L by CCR7

The U' equation will also be slightly modified without the $f_{dox}(t)$ term as we did not find any literature to indicate CCR7 performs the aforementioned functions only in the presence of another molecule. Furthermore, there is no evidence to suggest that PD-1 or PD-L1 is affected by CCR7 directly, so we can drop the $(1 + \lambda_{LU}U)$ factor from the L equation. We could also experiment with setting λ_{LU} to be extremely small in order to still apply the same form of the model.

5 Experimental Design

In order to assess how accurate the model is to the new tumor environment with the effects of CCR7 instead of UCP2, we design some potential experiments that could be used to further validate the model.

5.1 Xenograft Study

In order to study the impact CCR7 has on tumors, DCs, and CD8+ T cells, we propose a xenograft study of gastric cancer in a control mouse host as well as one in which the CCR7 gene has been completely knocked out. To justify an in vivo, xenograft approach, we note that because CCR7 is a cytokine receptor it is often difficult to stimulate it in a realistic fashion in vitro. Also, because some of the promotion of T-cells and DCs is via immigration into the TME, this is also not feasible to recreate in an in vitro environment. Furthermore, we choose a knockout gene setup instead of a knockdown because knocking out is not lethal to the mouse and has indeed been used in previous works [21]. The tumor growth, T-cell concentration, and DC concentration will be measured over several time points for both groups and compared. We hypothesize that in the CCR7 deficient case, there will be more tumor growth, and a lower T-cell and DC concentration.

5.2 ELISA CCR7 Measurement

In order to assess the CCR7 expression in M2 macrophages, we will have two groups of mice, one healthy and the other with gastric cancer. We will then separate the macrophages from the tumor environment from the diseased mice and also obtain the macrophages from the corresponding location in the healthy mice. After purification we then perform an enzyme-linked immunosorbent assay (ELISA) with a CCR7 kit in order to quantify the amount of CCR7 and compare between the healthy and tumorigenic mice. Our hypothesis is that there will be no significant difference between the two groups.

6 Future Direction

6.1 Evolutionary Algorithm

Initially, to fit the models quantitatively, we adapted Fu et. al.'s [11] evolutionary algorithm. We did not observe parameter convergence after around 5000 generations. In a future iteration of this project, we would increase the variance step to explore a wider range of the parameter space and run up to 15,000-20,000 generations.

6.2 Model Selection

Once we get the evolutionary algorithm to work, we can apply it to each of the three proposed models and use a model comparison test, such as akaike information criterion, in order to see which captures the data the best and to avoid over-fitting with the more complex models if possible.

6.3 Simulations with PD-1-PD-L1 Complex

Since the models incorporate the PD-1-PD-L1 complex, it would be useful to see the effects this could have on the microenvironment in combination with UCP2 and/or CCR7. Further in silico tests of this could be used to motivate future experimental work.

6.4 Further Mathematical Analysis

A key focus in this direction is to analyse the reduced dimensional model II analytically and study the stability of its equilibria. We intend to identify the presence of possible bifurcation phenomena and to illustrate them through simulations.

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